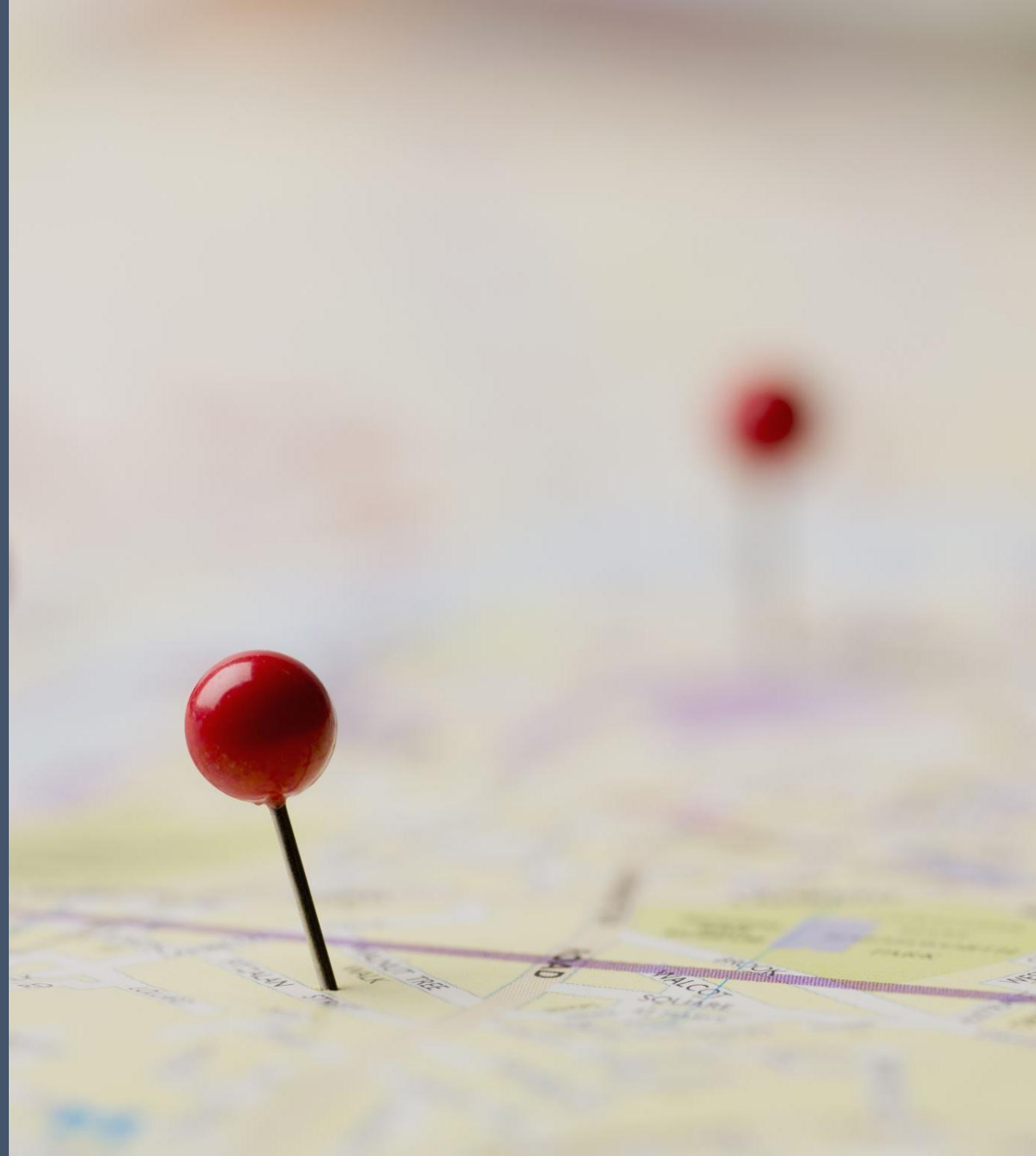
Building Trustworthy AI

Ethical AI: Laws, recommendations, policies

- https://ai-act-law.eu/recital/27/
- https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai
- https://www.europarl.europa.eu/RegData/etudes/BRIE/2019/640163/EPRS_BRI(2019)640163_EN.pdf
- https://artificialintelligenceact.eu/
- https://www.unesco.org/en/artificial-intelligence/recommendation-ethics
- https://github.com/EthicalML/awesome-artificial-intelligence-regulation



Ethical AI Priciples – Common Ground

- **Fairness and Bias Mitigation**
- **Transparency and "Explainability" (or interpretability)**
- **Privacy and Data Sovereignty**
- **Accountability and Human Oversight**
- **Safety and Reliability**
- **Social and Environmental Well-being**

The EU AI Act

- https://artificialintelligenceact.eu/ai-act-explorer/
- Definitions:
  - AI system
  - Risk
  - Reasonably foreseeable misuse
  - Serious incident
  - Deep fake
  - Systemic risk
  - Prohibited Ai practices (Article 5)
  - Transparency Obligations (Article 50)
  - …

Why?

- **Global MOOC on the Ethics of AI:**
  - **Module 1 Video 1: Why does ethics matter in AI?**
    - https://www.coursera.org/learn/global-mooc-on-the-ethics-of-ai/lecture/jRdAH/video-1-why-does-ethics-matter-in-ai
  - **Module 2 Video 2: Humanization as a design imperative - design for flourishing**
    - https://www.coursera.org/learn/global-mooc-on-the-ethics-of-ai/lecture/jePSO/video-2-humanization-as-a-design-imperative-design-for-flourishing
  - **Module 2 End-of-Module Exercise**

Glossary

- **Ethics:** The systematic study of what matters, what is right, and how we justify choices — not a fixed rulebook.
- **Normative ethics:** The branch of ethics that asks, "what should we do?" and generates principles for action.
- **Applied ethics:** Translating ethical principles into real decisions in specific domains such as medicine, law, journalism, and AI.
- **Utilitarianism:** A framework that judges actions by their outcomes; applied to AI, it favours deploying a system if net welfare is positive.
- **Deontology:** A framework that judges actions by duties and rights regardless of outcomes, holding that some protections (dignity, due process, non-discrimination) are non-negotiable.
- **Virtue ethics:** A framework asking what a person or institution of good character would do — what kind of organisation we become through our design choices.
- **Ubuntu:** An African relational ethical concept — "a person is a person through other persons" — that shifts the question from individual consent to communal impact.
- **Communitarian ethics:** An approach treating the community, rather than the individual, as the primary unit of moral concern.
- **Confucian role ethics:** A tradition holding that moral character is formed through practice and relationship, implying that no design culture is morally neutral.
- **Buddhist ethics:** A tradition centring non-harm and interdependence — a model sits within a network of labellers, users, and downstream consequences rather than being a discrete artefact.
- **Indian dharmic traditions:** A tradition treating duty as context-sensitive — what is right depends on role, situation, and time, sometimes requiring responsible choice among imperfect options.
- **Ethical pluralism:** The condition in which the same facts yield several defensible ethical verdicts because different traditions optimise different values.
- **Five-question framework:** A diagnostic lens applied to any AI system: Who benefits? Who is harmed? Who decides? Who can challenge? At what scale?
- **Hallucination:** When an AI system produces confident misinformation without reliable awareness of its own errors.
- **Parasocial dependency:** An emotional attachment a user forms toward a non-human system that does not meaningfully reciprocate.
- **Algorithmic bias:** When a model learns historical inequities from training data and replicates them as discriminatory outputs at scale.
- **Transparency:** Openness about how a system works, what data it uses, and where it fails — a precondition for accountability, but not identical to it.
- **Explainability:** The property of a system whose decisions can be understood or traced; useful, but not a substitute for accountability.
- **Accountability:** The ability to identify and challenge those responsible for a system's decisions.
- **Ethics by design:** Embedding values into a system before it is built or deployed — across purpose, data, model architecture, and governance gates.
- **Ethics in use:** Monitoring, correcting, and remaining accountable once a system is deployed, feeding what is learned back into the next design cycle.
- **Humanization (over automation):** A design principle centring empathy, accountability, and cultural diversity so systems support whole persons rather than replacing human judgment and connection.

Do not harm – who can be harmed and why?

- Safety vs Security – Exercise – what is the difference?

  - Are non-users affected at all?

- This is the "uncomfortable truth" of AI: unlike traditional software that crashes, AI failures are often invisible. These silent biases can affect thousands of decisions over months or years before anyone notices.

Case study

Autonomous Driving: Promise vs. Reality

THE PROMISE

- Eliminate 94% of crashes caused by human error (~1.35M lives saved annually)
- Provide mobility to elderly and disabled populations
- Reduce congestion through optimized routing
- Transportation deaths become rare exceptions, not daily tragedies

THE REALITY

- Algorithmic life-or-death decisions
- Unpredictable failures
- Mixed-traffic danger
- Cybersecurity vulnerabilities
- Software update risks
- Sensor limitations
- No fail-safe

Involve all parties into the design - why

Who is affected?

Stakeholder	Impact	Choice
The victim	Fatal injury	*No choice—involuntary participant*
Other pedestrians & drivers	Exposed to deployment risks	*No choice—involuntary participants*
Passengers	Safety risks, but also service benefits	*Limited choice—imperfect information*
Safety driver	Faced criminal charges, trauma	*Employment choice, but complex pressures*
Uber	Regulatory scrutiny	*Full decision-making power*

Involve all parties into the design - why

Three launch options

Question: What level of risk to non-consenting pedestrians and other drivers is acceptable during the learning phase?

Approach	Core Trade-Off	Risk Allocation
A: Aggressive Deployment	Speed to market vs. safety validation	Risk shifted to drivers (with disclaimers)
B: Extended Testing	Market delay vs. comprehensive safety	Risk retained by company during long validation
C: Geofenced Rollout	Limited market vs. controlled environment	Phased risk—lower exposure but slower expansion

Underlying question:
What level of risk to non-consenting pedestrians and drivers is acceptable during the learning phase?

Moving forward: What we learned



1. Rigorous Pre-Deployment Testing

Question: Not 'has it been tested?' but '**tested adequately** for specific risks?'



2. Design Choices Encode Values

Reducing false positives = **comfort over safety**—an ethical decision, not just technical



3. Post-Incident Transparency

Company responses shape **public trust, regulation, and industry learning**



4. Regulatory Frameworks Matter

Absence of standards ≠ freedom to innovate. It means **responsibility ambiguity** serving neither companies nor public



5. Multi-Stakeholder Governance

Engineers, regulators, affected communities, civil society need seats at the table

3 commonly used, but interlinked terms - Testing, Evaluation, and Benchmarking

Term	What it means	Main question
Testing	Checking how the system behaves under specific conditions, including edge cases and failures	Does it work safely and reliably in this scenario?
Evaluation	Assessing overall performance, risks, and fitness for purpose using technical and non-technical criteria	Is this system good enough and appropriate to deploy or use?
Benchmarking	Comparing a system against standardized datasets, tasks, or other models	How does it compare to a baseline or peer system?

Why AI Needs a Testing Framework



AI is Not Like Other Software

AI systems learn from data, but their outputs shift across contexts

▼

AI testing goes beyond standard QA



Harm Happens at Scale

AI systems can reach millions of users simultaneously, affecting everyone in scope

▼

AI testing accounts for harm at scale



Accountability Has No Hook

When AI causes harm, responsibility is diffusedacross developers, deployers, and regulators

▼

AI testing helps create clear standards of accountability

Testing as the first step to regulation

What we look for?

How many scenarios were tested?
Volume and breadth of test coverage

What conditions were validated?
Edge cases, populations, environments

Which populations were included?
Demographic diversity, equity

Regulation

Scope and Risk Classification
Classification, obligation, and documentation

Declaration of Conformity
Formal sign-off that key requirements are met

Post-Market Monitoring
Tracking, Incident reporting, and updates

Evaluation as the Next Step To Regulation

What we look for?		Regulation
What outcomes were evaluated? Performance, harms, reliability	→	**Risk Thresholds** Minimum performance and safety levels
What impacts were assessed? Fairness, privacy, human oversight	→	**Safeguards** Controls matched to identified risks
What evidence supports deployment? Reports, documentation, review	→	**Assurance & Monitoring** Audit readiness, tracking, updates

Tool #1 : Datasheets for Datasets

The Core Questions a Datasheet Answers:

1. Motivation:
Why was this dataset created? Who funded it?

2. Composition:
What do the instances represent (images, text, people)?

3. Collection:
How was the data acquired? Was it scraped, or was there consent?

4. Preprocessing:
Was the data cleaned, normalized, or anonymized?

5. Distribution:
Are there any copyrights or usage restrictions?

Many organizations now provide these as a standard part of their "Safety and Security" stack.

Examples of Companies & Services using Datasheets

Organization	Implementation / Service
Microsoft	Transparency Notes
Hugging Face	Dataset Cards
Google	Data Cards Playbook
IBM	AI FactSheets
Common Crawl	Data Governance Docs

Tool #2: Model Cards - Definition and Importance

A **Model Card** is a standardized, brief document that provides a structured overview of a machine learning model's purpose, design, and performance.

Why they are critical for Safety & Security?



Risk Management



Accountability



Security Posture



Regulatory Compliance

Tool #2: Model Cards - Standard Content



Model Details
Model name, version, architecture (e.g., Transformer), and developer contact info.

Intended Use
Primary tasks (e.g., sentiment analysis) and "out-of-scope" uses (e.g., "not for clinical diagnosis").

Factors & Demographics
Subgroups the model was tested on (race, gender, age) to highlight performance gaps.

Metrics & Thresholds
Accuracy, F1-score, or False Positive rates, including the decision thresholds used.

Training & Eval Data
Links to the Datasheets for the sets used to train and validate the model..

Quantitative Analysis
A confusion matrice will identify specific types of errors..

Safety Disclosures
Results from safety testing, vulnerability assessments, and mitigation steps taken.

Tool #2: Model Cards - Roles and Examples

Company	Implementation Name
Google	Gemma Model Card
OpenAI	System Cards (GPT-4o)
Salesforce	AI Transparency Notes
NVIDIA	Model Card++
Hugging Face	Standard Model Cards